Decoder's Book

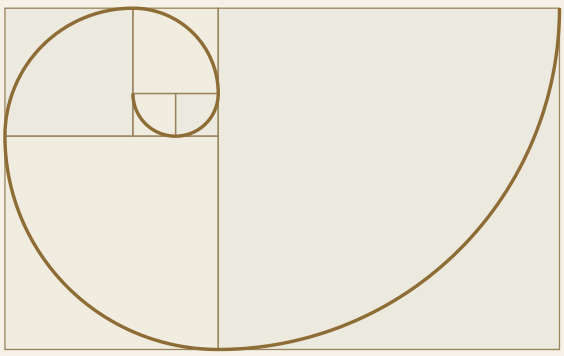
CERTIFICATE OF DISCOVERY

01

The Pattern Keeper

Fibonacci

The hidden order of growth



For identifying the Fibonacci sequence in the plant's growth and decoding the idea hidden in the garden.

OBSERVATION · COUNTING · REASONING

PUZZLE SOLVED

The Moonlit Garden

ANSWER RECOGNISED

FIBO / FIBONACCI

A chapter earned by solving a puzzle. An idea you can take with you.

What the plant was hiding

Each watering adds a branch and a new group of leaves. The clue is the number of leaves added by that watering, rather than the total number of leaves on the plant.

01	02	03	04	05
1	1	2	3	5

WATERING	NEW LEAVES	TOTAL LEAVES
1	1	1
2	1	2
3	2	4
4	3	7
5	5	12

01 **Observe**
Water the plant five times and watch each new group of leaves appear.

02 **Count**
Record the new leaves in order: 1, 1, 2, 3, 5.

03 **Connect**
From the third number onwards, each is the sum of the previous two.

04 **Decode**
Recognise the pattern's name and enter FIBO or FIBONACCI.

> ### The distinction that matters
> After five waterings there are 12 leaves: 1 + 1 + 2 + 3 + 5. Twelve is the total, not the next term of the sequence. The next group would contain 8 leaves.

You turned a visual event into a numerical rule, then the rule into a name.

THE IDEA BEHIND THE PUZZLE

Two numbers make the next

Start with two values and keep adding the latest pair.

$$F_n = F_{n-1} + F_{n-2}$$

The usual definition starts with $F_0 = 0$ and $F_1 = 1$. For $n \geq 2$, $F_n = F_{n-1} + F_{n-2}$. The puzzle uses the positive terms, starting with 1 and 1. [2]

THE FIRST TERMS

0 · 1 · 1 · 2 · 3 · 5 · 8 · 13 · 21 · 34 · 55

1 + 1 = 2 1 + 2 = 3 2 + 3 = 5 3 + 5 = 8

Who was Fibonacci?

Leonardo of Pisa, known as Fibonacci, was a medieval mathematician born around 1170. His Liber abaci, completed in 1202, explained calculation with Indo-Arabic numerals. It includes an idealised rabbit-breeding problem now associated with the sequence named after him. The pattern had also been studied by Indian mathematicians before his work. [1] [2]

Generate it yourself

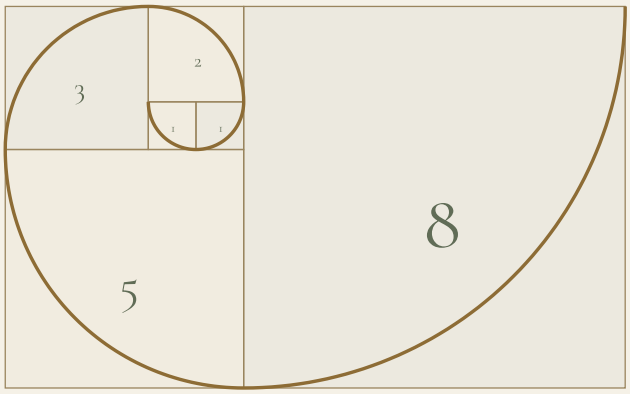
Keep just two numbers. Write the first, then replace the pair with the second number and their sum. Repeat.

```
a = 0, b = 1
repeat:
    write a
    a, b = b, a + b
```

In the last line both values are updated using the old pair.

> The two starting values are part of the definition; that is why there are two ones. The addition rule begins after them.

A pattern you can draw



Squares with sides 1, 1, 2, 3, 5 and 8, placed beside one another.

The arcs are quarter-circles, one in each square. This is an approximate Fibonacci spiral; the exact golden spiral is a logarithmic curve, rather than a chain of circular arcs. [3]

The link to the golden ratio

Ratios of consecutive terms approach $\varphi = (1 + \sqrt{5}) / 2 \approx 1.618034$. Early ratios alternate around it; they are not all equal to φ. [2]

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5/3 → 8/5 → 13/8 → 21/13
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A model, rather than a rule for every plant

In this puzzle the groups of leaves were deliberately designed as 1, 1, 2, 3, 5. Recognising the pattern does not prove that every plant grows this way, or that every natural spiral is a golden spiral.

Check your understanding

What follows 1, 1, 2, 3, 5?
8, 13, 21. Each time, add the latest two values.

If the puzzle continued with a sixth watering?
There would be 8 new leaves and 20 leaves in total.

Further reading

1. ETH Library · Leonardo of Pisa
2. Wolfram MathWorld · Fibonacci Number
3. Wolfram MathWorld · Golden Spiral